

SUPPLEMENT TO CERTIFIED BELLMAN OPERATORS FOR COSTLY POLICY REVISION

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1. READING CONVENTIONS AND PROVENANCE

This supplement accompanies revision R45. All numerical optimization outcomes below are the preserved R43 development experiments or the R44 frozen global experiment. R45 contributes their integrated analysis, independent rechecking, new analytic results, and exact tests of the stopping contract. It does not rebrand an already observed experiment as a new prospective sample.

B denotes Bellman supports with disaggregated simplex-consistent products. A denotes the aggregate-product relaxation. A certified intersection takes the larger of the two independently checked lower bounds and the smallest independently checked feasible upper cost, also allowing the rationally repaired SCIP proposal. SCIP's native lower bound is not used in that intersection. Absolute widths are compared with the registered target 10^{-3} . Rounded table endpoints are descriptive; the exact fractions in the machine-readable results and proof objects determine validity and target status.

The global cases are twelve designed environments (four each in maintenance, inventory, and queueing) plus four perturbations of one near-tie construction. Forty-two additional scaling outputs are three model families, seven allowances, and two root formulations. They are not forty-two independent environments. The historical controlled suite consists of eight cases and two root formulations. Its fiber calculation is one parametric family with seventeen endpoints.

1.1. *Designed family primitives*

Inventory has state $i \in \{0, \dots, n-1\}$, action $a \in \{0, \dots, m-1\}$, and demand shock $z \in \{0, 1, 2\}$. Set $o = \min\{a, n-1-i\}$, $b = i + o$, $s = \min\{b, z\}$, and $i' = b - s$. The stage operating payoff is $2s/n - 3o/(4n) - i'/(5n) - (z-b)_+/(2n)$. The installed action is $\min\{m-1, \max\{0, n/2-i\}\}$ for the executed even state counts, the departure charge is $(1+i/n)\mathbf{1}\{a \neq a^0(i)\}$, and terminal reward is $i/(4n)$.

Queueing uses $i' = \min\{n-1, (i-a)_+ + z\}$ and stage operating payoff $-i/n - a^2/(3m^2) - (i-a+z-(n-1))_+/n$. Its installed action is one below $n/2$ and $\min\{2, m-1\}$ otherwise; departure costs are again $1+i/n$, and terminal reward is $-i/(2n)$. In both families the initial regime is uniform. Shock probabilities are rational $(w_0, w_1, 8-w_0-w_1)/8$ with $w_0 \in \{1, 2\}$ and $w_1 \in \{3, 4\}$ chosen by the frozen seed. Maintenance uses the preserved controlled rational transition constructor and departure charges from R43; every resulting primitive is serialized in the frozen model files. All three families are known-model designs, not estimated data.

2. FULL GLOBAL RESULTS

TABLE I: Best independently certified intersections

Case	Lower	Upper	Width	Width/upper (%)	Target
M0	1.2487	1.2491	0.000497	0.0398	Yes
M1	4.8670	4.8741	0.007157	0.1468	No
M2	10.6217	10.6304	0.008723	0.0821	No
M3	16.2520	16.3659	0.1139	0.6957	No
I0	0	0	0	0	Yes
I1	4.5308	4.5486	0.0179	0.3925	No
I2	11.2217	11.2359	0.0142	0.1263	No
I3	15.5942	15.5988	0.004630	0.0297	No
Q0	0.6363	0.6363	6.129×10^{-12}	0.0000	Yes
Q1	3.7682	3.7692	0.000980	0.0260	Yes
Q2	7.5909	7.5920	0.001151	0.0152	No
Q3	11.6764	11.6771	0.000660	0.0057	Yes
T0	5.2227	5.2277	0.004995	0.0956	No
T1	5.2224	5.2277	0.005256	0.1005	No
T2	5.2272	5.2305	0.003211	0.0614	No
T3	8.3022	8.4367	0.1345	1.5941	No

TABLE II: Covering-search progress and stopping events

Case	Method	Nodes	Stop	Root width	Final width
M0	B	1	target	0.000497	0.000497
M0	A	13	target	0.005303	0.000567
M1	B	363	time_cap	0.015462	0.007157
M1	A	1125	time_cap	0.055623	0.022168
M2	B	45	time_cap	0.010616	0.008723
M2	A	209	time_cap	0.033659	0.025514
M3	B	7	time_cap	0.728567	0.728567
M3	A	13	time_cap	0.115013	0.115011
I0	B	1	target	0	0
I0	A	1	target	0	0
I1	B	239	time_cap	0.030765	0.017855
I1	A	911	time_cap	0.058425	0.026761
I2	B	25	time_cap	0.014884	0.014188
I2	A	209	time_cap	0.047759	0.034881
I3	B	7	time_cap	0.064000	0.064000
I3	A	13	time_cap	0.005123	0.004768
Q0	B	1	target	7.607×10^{-12}	7.607×10^{-12}
Q0	A	1	target	6.129×10^{-12}	6.129×10^{-12}
Q1	B	49	target	0.002272	0.000980
Q1	A	153	target	0.006964	0.000997
Q2	B	81	time_cap	0.002249	0.001151
Q2	A	331	time_cap	0.007350	0.004986
Q3	B	7	time_cap	0.032158	0.032158
Q3	A	1	target	0.000694	0.000694
T0	B	879	time_cap	0.029761	0.004995

Case	Method	Nodes	Stop	Root width	Final width
T0	A	2047	node_cap	0.210087	0.160574
T1	B	789	time_cap	0.029761	0.005256
T1	A	2047	node_cap	0.210088	0.200032
T2	B	879	time_cap	0.029263	0.003211
T2	A	2047	node_cap	0.212848	0.167849
T3	B	1053	time_cap	0.176091	0.134494
T3	A	2047	node_cap	3.004802	1.360331

A target stop certifies the stated numerical tolerance, not exact equality of the two endpoints. A time or node stop preserves the surviving global cover. In particular, no numerical LP infeasibility status alone is allowed to discard a policy box. Q1 closes after 49 B nodes and 153 A nodes. M0 closes at B's root and after 13 A nodes. Several target hits, including the zero-cost I0 example, occur at the root. The largest cases in all three primary families favor A under the common budget.

TABLE III: Search, LP, and independently checking work

Case	Method	Search (s)	LP (s)	Check (s)	Proof (KiB)
M0	B	0.100	0.016	0.077	2.7
M0	A	0.367	0.080	0.237	6.4
M1	B	120.396	40.424	80.470	4005.7
M1	A	120.223	23.048	77.318	2592.6
M2	B	120.178	83.605	34.056	2189.0
M2	A	120.679	46.165	58.524	7093.8
M3	B	128.174	87.564	27.164	2105.0
M3	A	124.350	90.002	18.922	4182.9
I0	B	0.082	0.012	0.065	0.8
I0	A	0.028	0.005	0.018	0.7
I1	B	120.582	53.186	69.526	1217.5
I1	A	120.108	27.079	74.702	1821.5
I2	B	120.592	96.763	20.994	352.6
I2	A	120.132	55.471	50.890	1689.6
I3	B	126.244	92.112	25.605	187.4
I3	A	120.886	94.694	13.401	568.2
Q0	B	0.067	0.011	0.054	1.4
Q0	A	0.021	0.004	0.012	1.3
Q1	B	25.720	11.935	13.877	272.0
Q1	A	18.778	3.902	11.701	206.2
Q2	B	121.181	69.953	51.340	519.5
Q2	A	120.178	51.433	53.891	673.5
Q3	B	127.730	85.776	30.474	51.5
Q3	A	16.055	7.098	1.456	18.3
T0	B	120.073	39.233	79.651	1309.3
T0	A	105.073	25.732	62.810	2957.8
T1	B	120.186	43.859	74.915	1263.4
T1	A	100.177	21.173	61.847	833.4
T2	B	120.067	43.136	75.872	1326.2
T2	A	100.465	23.915	59.707	2722.9

Case	Method	Search (s)	LP (s)	Check (s)	Proof (KiB)
T3	B	120.118	53.741	64.360	1582.6
T3	A	70.653	18.852	40.171	3322.7

The times above are the original per-case execution records, not a new benchmark on the present machine. Search includes non-LP preparation and exact node operations. Check time is additional. The nominal time budget is tested between proof-producing operations, so completion of an operation or sibling pair can exceed it. The JSON records retain the complete trace and measured totals.

TABLE IV: Global root proposal status

Case	Method	LP status	Primal	Dual	Variables
M0	B	0	Yes	Yes	296
M0	A	0	Yes	Yes	160
M1	B	0	Yes	Yes	1496
M1	A	0	Yes	Yes	704
M2	B	0	Yes	Yes	4128
M2	A	0	Yes	Yes	2080
M3	B	1	No	No	17952
M3	A	0	Yes	Yes	8512
I0	B	0	Yes	Yes	278
I0	A	0	Yes	Yes	136
I1	B	0	Yes	Yes	1580
I1	A	0	Yes	Yes	648
I2	B	0	Yes	Yes	5088
I2	A	0	Yes	Yes	1900
I3	B	1	No	No	23656
I3	A	0	Yes	Yes	8140
Q0	B	0	Yes	Yes	278
Q0	A	0	Yes	Yes	124
Q1	B	0	Yes	Yes	1580
Q1	A	0	Yes	Yes	620
Q2	B	0	Yes	Yes	5088
Q2	A	0	Yes	Yes	1720
Q3	B	1	No	No	23904
Q3	A	0	Yes	Yes	7768
T0	B	0	Yes	Yes	538
T0	A	0	Yes	Yes	324
T1	B	0	Yes	Yes	538
T1	A	0	Yes	Yes	324
T2	B	0	Yes	Yes	538
T2	A	0	Yes	Yes	324
T3	B	0	Yes	Yes	538
T3	A	0	Yes	Yes	324

Status zero denotes a returned successful numerical LP proposal; it does not certify the nonlinear problem. The full tree contains the rounded multipliers and their reconstructed outward

residual corrections. The independent checker rebuilds their contribution to each node lower bound. No missing optimizer-specific scalar is invented to replace that calculation.

TABLE V: External SCIP baseline and the separate certified intersection

Case	SCIP status	Native width	Nodes	Total (s)	Certified width
M0	gaplimit	0.000497	1	0.11	0.000497
M1	gaplimit	0.000832	11	3.86	0.007157
M2	gaplimit	0.000835	1	12.96	0.008723
M3	timelimit	0.0520	1	120.52	0.1139
I0	optimal	0	0	0.05	0
I1	nodelimit	0.001326	2047	14.26	0.0179
I2	nodelimit	0.003895	2047	72.75	0.0142
I3	timelimit	0.001470	1	120.43	0.004630
Q0	gaplimit	7.000×10^{-9}	1	0.06	6.129×10^{-12}
Q1	gaplimit	0.000243	1	1.35	0.000980
Q2	gaplimit	2.670×10^{-5}	1	4.98	0.001151
Q3	gaplimit	0.000149	1	13.14	0.000660
T0	nodelimit	0.002574	2047	6.55	0.004995
T1	timelimit	0.002388	1359	120.01	0.005256
T2	nodelimit	0.002624	2047	6.17	0.003211
T3	nodelimit	0.1092	2047	10.85	0.1345

SCIP’s native lower endpoints are solver-reported floating-point quantities. The native target count is eight; the independently certified intersection count is five. Every SCIP proposed upper policy is separately rationalized, repaired, and checked at all restarts. These distinct checks are not a certification of SCIP’s native lower endpoint. The native solver often supplies tighter bounds: the result is retained rather than filtered out. The adapter gives SCIP the same common-policy equations, admissible root domains, Bellman supports, and valid strict-gap cuts.

3. ACTION GAPS, SUPPORT CONSTANTS, AND ARITHMETIC PRECISION

TABLE VI: Constants across all global cases

Case	Strict	$\underline{d}$	λ_-	λ_+	Exact-LP bound
M0	Yes	0.002320	-190.5950	767.8502	16.1024
M1	Yes	0.003793	-166.1942	183.0214	54.5307
M2	Yes	0.001033	-111.7540	1660.1008	162.9533
M3	Yes	5.146×10^{-5}	-25440.8333	1774.2602	420.1714
I0	No	0	-19.3268	0	–
I1	No	0	-37.9259	24.4222	–
I2	No	0	-19.6491	25.1542	–
I3	No	0	-42.1053	47.3684	–
Q0	Yes	0.001389	-1080.0000	27.0000	4.7987
Q1	Yes	0.0223	-61.5544	69.1200	18.1096
Q2	Yes	0.003125	-440.0000	560.0000	134.2563

Case	Strict	$\underline{d}$	λ_-	λ_+	Exact-LP bound
Q3	Yes	0.008854	-162.3529	94.4762	35.9911
T0	No	0	0	8.0902	—
T1	Yes	1.000×10^{-8}	0	9.953×10^7	1.426×10^{12}
T2	Yes	0.000100	0	9952.9688	14318.5489
T3	Yes	0.0100	0	99.5297	49.7421

A dash is not a zero bound. If an operating tie exists, the inverse-gap theorem is inapplicable. The ratio extrema shown in those rows describe the nonzero disadvantages only and do not establish a cone for the zero-disadvantage actions. Bellman support envelopes and the policy-box stopping theorem do not require strict separation. The first-index reference is computed in exact rational arithmetic for the stored primitives. Its identity can change at a tie; the reference-free support expression remains valid on either side.

TABLE VII: Exact-repair precision contract evaluated on the frozen models

Case	D_C	D_J	$H(0, 44)$	Width level k	Depth bound
I0	14.255500	11.675775	2.787×10^{-9}	29	7424
I1	53.408547	17.098336	4.087×10^{-8}	32	32768
I2	155.828997	40.479846	3.967×10^{-7}	37	113664
I3	384.671359	71.774254	7.052×10^{-6}	41	503808
M0	16.292000	18.916400	3.185×10^{-9}	30	7680
M1	56.969117	62.276480	4.360×10^{-8}	34	34816
M2	166.217597	173.259624	4.232×10^{-7}	39	119808
M3	397.080112	403.278880	7.280×10^{-6}	44	540672
Q0	14.255500	18.260300	2.787×10^{-9}	29	7424
Q1	53.408547	62.931790	4.087×10^{-8}	34	34816
Q2	155.828997	182.767970	3.967×10^{-7}	39	119808
Q3	384.671359	452.526471	7.052×10^{-6}	44	540672
T0	49.847977	15.666507	3.815×10^{-8}	32	16384
T1	49.847977	15.666507	3.815×10^{-8}	32	16384
T2	49.847977	15.666507	3.815×10^{-8}	32	16384
T3	49.847977	15.666507	3.815×10^{-8}	32	16384

Here $q = 2^{-44}$, the arithmetic accumulation is $\gamma = q \sum_{s=0}^{T-1} \beta^s$, and H is the stopping bound in the main article. The width level means $w = 2^{-k}$, sufficient for $H(w, 44) \leq 10^{-3}$ using the displayed model bounds. The depth bound is $8nT(m-1)k$, not an observed tree size or an efficient complexity guarantee. These are exact-repair floors. They do not by themselves charge the frozen probability grid or its repair guard; the separate table below does. A smaller arbitrary tolerance can require increasing both precisions. The exact fractions and inequalities are in `precision_contract.json`.

The independent modulus test constructs 256 rational boxes in a two-state, three-date, three-action model with an exact operating tie. It separately evaluates restricted Bellman minima, repairs the lower-corner policy, and verifies the stopping inequality with exact fractions. All 256 survived the necessary tests and passed. This is finite property testing, not a machine-checked proof of the theorem. The mathematical argument is given in the main article’s Appendix A.

TABLE VIII: The implemented two-grid contract, including reference-repair rounding

Case	$\max d$	$\mathcal{H}_Q(0, 44, 40)$	Prob. bits	Width level	Depth bound
I0	0.558594	5.343×10^{-8}	40	29	7424
I1	0.265971	4.013×10^{-7}	40	32	32768
I2	0.394211	7.995×10^{-6}	40	37	113664
I3	0.385802	0.000138	40	41	503808
M0	0.285941	3.327×10^{-8}	40	30	7680
M1	0.400005	6.150×10^{-7}	40	34	34816
M2	0.527327	1.123×10^{-5}	40	39	119808
M3	0.542239	0.000197	40	44	540672
Q0	0.629561	5.976×10^{-8}	40	29	7424
Q1	1.064878	1.446×10^{-6}	40	34	34816
Q2	0.932956	1.825×10^{-5}	40	39	119808
Q3	1.075412	0.000371	40	44	540672
T0	0.275000	3.855×10^{-7}	40	32	16384
T1	0.275000	3.855×10^{-7}	40	32	16384
T2	0.275000	3.855×10^{-7}	40	32	16384
T3	0.275000	3.855×10^{-7}	40	32	16384

The frozen node-bound precision is 44 bits, while the inherited repair routine floors nonreference action probabilities at 40 bits and assigns the residual mass to the operating reference. Its exact guard is charged in Corollary 1 of the main article. All sixteen models satisfy the corollary’s positive-slack condition and have sufficient two-grid floors below 10^{-3} ; the largest is approximately 0.0003714041. The displayed width levels refer to this quantized repair, not to the ideal repair. The worst-case depth is an analytic bound, not executed work. A second exact implementation verified this contract on the same 256 boxes used for the ideal-repair test, at several value and probability precisions. No adaptive-precision search was executed.

4. COMPLETE ALLOWANCE DIAGNOSTICS

TABLE IX: Seven allowances in three families, with both root formulations

Family	ε	Unscaled width	Scaled width	Scaled/ ε^2	Exact-LP bound
M	10^{-1}	1.8633	1.8633	186.3299	1149.4310
M	10^{-2}	0.9721	0.9567	9567.2260	64.9860
M	10^{-3}	0.1961	0.0140	14048.4194	34.2205
M	10^{-4}	0.0187	0.000144	14406.2818	0.8230
M	10^{-5}	0.001858	1.444×10^{-6}	14444.6438	0.009579
M	10^{-6}	0.000186	2.113×10^{-8}	21125.6740	9.739×10^{-5}
M	10^{-7}	1.857×10^{-5}	8.648×10^{-10}	86480.7821	9.755×10^{-7}
I	10^{-1}	2.6080	2.6080	260.8030	—
I	10^{-2}	0.2168	0.2168	2168.1330	—
I	10^{-3}	0.0212	0.0212	21213.1715	—
I	10^{-4}	0.002117	0.002117	2.117×10^5	—
I	10^{-5}	0.000212	0.000212	2.116×10^6	—
I	10^{-6}	2.116×10^{-5}	2.116×10^{-5}	2.116×10^7	—
I	10^{-7}	2.116×10^{-6}	2.116×10^{-6}	2.116×10^8	—

Family	ε	Unscaled width	Scaled width	Scaled/ ε^2	Exact-LP bound
Q	10^{-1}	0.6245	0.1050	10.5035	48.3406
Q	10^{-2}	0.0444	1.259×10^{-10}	1.259×10^{-6}	8.0827
Q	10^{-3}	0.004533	1.156×10^{-10}	0.000116	0.1854
Q	10^{-4}	0.000453	1.197×10^{-10}	0.0120	0.002130
Q	10^{-5}	4.525×10^{-5}	1.217×10^{-10}	1.2172	2.162×10^{-5}
Q	10^{-6}	5.3153	7.825×10^{-10}	782.5364	2.165×10^{-7}
Q	10^{-7}	4.527×10^{-7}	1.354×10^{-10}	13540.7430	2.166×10^{-9}

TABLE X: Complete strict-gap bound components for the scaled formulation

Family	ε	$\bar{E}_D$	$E_{S,0}$	D_C	χ	ρ	Repair term
M	10^{-1}	0.9286	1092.7670	56.9691	1.0000	0.9946	56.6640
M	10^{-2}	0.009286	10.9277	56.9691	1.0000	0.9489	54.0583
M	10^{-3}	9.286×10^{-5}	0.1093	56.9691	0.9212	0.6500	34.1112
M	10^{-4}	9.286×10^{-7}	0.001093	56.9691	0.0921	0.1566	0.8220
M	10^{-5}	9.286×10^{-9}	1.093×10^{-5}	56.9691	0.009212	0.0182	0.009568
M	10^{-6}	9.286×10^{-11}	1.093×10^{-7}	56.9691	0.000921	0.001854	9.728×10^{-5}
M	10^{-7}	9.286×10^{-13}	1.093×10^{-9}	56.9691	9.212×10^{-5}	0.000186	9.744×10^{-7}
I	10^{-1}	—	—	—	—	—	—
I	10^{-2}	—	—	—	—	—	—
I	10^{-3}	—	—	—	—	—	—
I	10^{-4}	—	—	—	—	—	—
I	10^{-5}	—	—	—	—	—	—
I	10^{-6}	—	—	—	—	—	—
I	10^{-7}	—	—	—	—	—	—
Q	10^{-1}	0.0843	3.5231	47.4743	1.0000	0.9440	44.8175
Q	10^{-2}	0.000843	0.0352	47.4743	0.2700	0.6278	8.0475
Q	10^{-3}	8.435×10^{-6}	0.000352	47.4743	0.0270	0.1443	0.1850
Q	10^{-4}	8.435×10^{-8}	3.523×10^{-6}	47.4743	0.002700	0.0166	0.002126
Q	10^{-5}	8.435×10^{-10}	3.523×10^{-8}	47.4743	0.000270	0.001684	2.159×10^{-5}
Q	10^{-6}	8.435×10^{-12}	3.523×10^{-10}	47.4743	2.700×10^{-5}	0.000169	2.162×10^{-7}
Q	10^{-7}	8.435×10^{-14}	3.523×10^{-12}	47.4743	2.700×10^{-6}	1.687×10^{-5}	2.162×10^{-9}

The clean bound equals $E_{S,0} + D_C \chi \rho$ with $\rho = \bar{E}_D / ((1 - \beta)\varepsilon + \bar{E}_D)$. It describes an exactly optimizing relaxed solution and exact repair. The published numerical interval additionally reflects proposal suboptimality, directed rounding, dual-residual corrections, and candidate construction. These are not generally identified by the numerical LP status. An independently certified numerical LP optimality-loss scalar is not available for these records; it is left unquantified, not set to zero. The exact residual construction still establishes each reported endpoint directly.

The inventory models contain ties, so the strict-gap theorem does not apply and both root formulations coincide in the small-allowance sequence. Maintenance exhibits a roughly quadratic intermediate range followed by arithmetic effects. Queueing exhibits a small-width floor and an unfavorable unscaled endpoint at allowance 10^{-6} . All points are retained. Descriptive secants appear in the JSON, but no common asymptotic exponent is estimated from these different regimes.

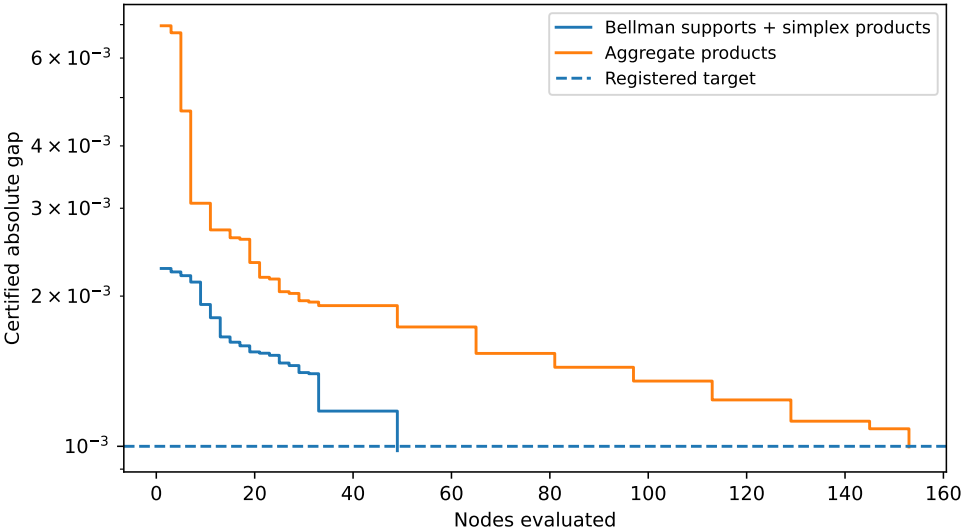


FIGURE 1.—Q1: full-cover certified gap versus nodes. The main article reports the corresponding time trace.

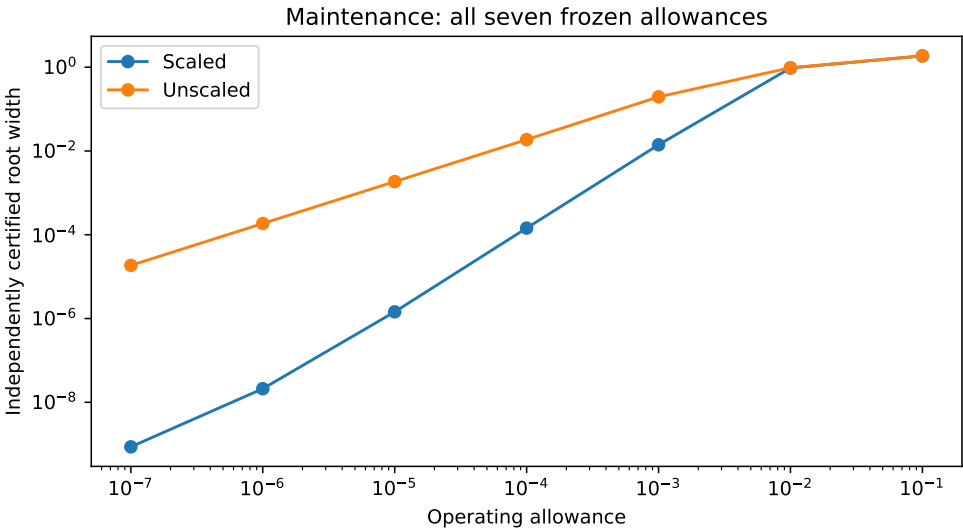


FIGURE 2.—Maintenance: every frozen allowance point. These are measured certificate widths, not the exact-LP theorem bound.

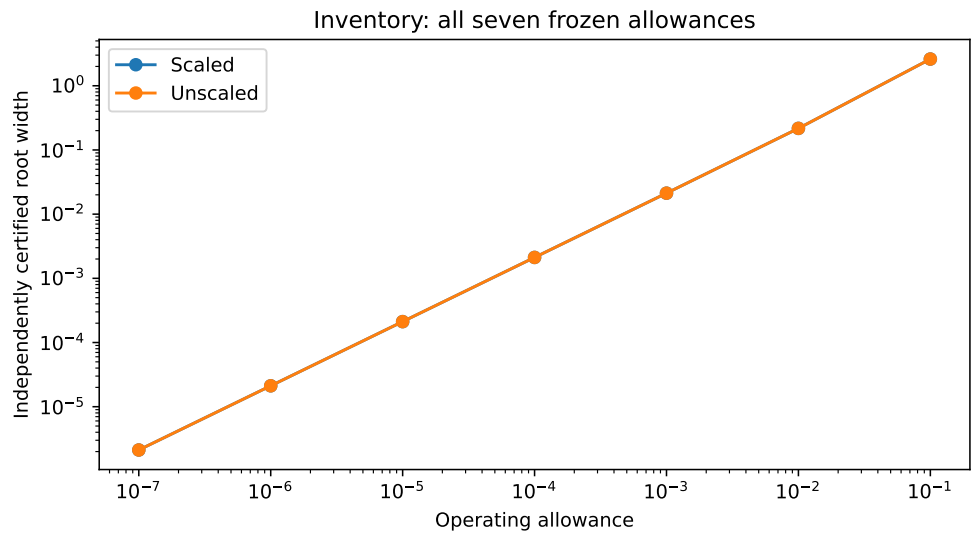


FIGURE 3.—Inventory: every frozen allowance point. These are measured certificate widths, not the exact-LP theorem bound.

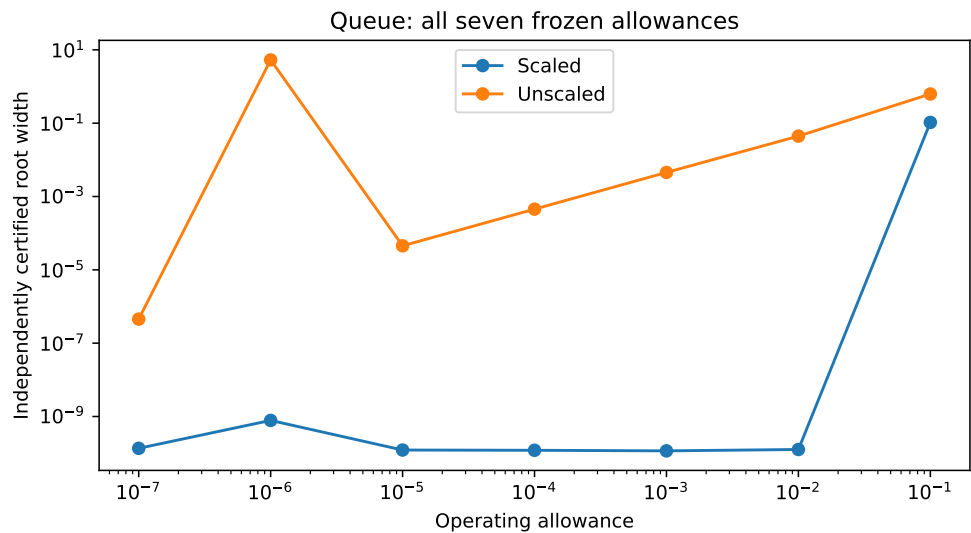


FIGURE 4.—Queue: every frozen allowance point. These are measured certificate widths, not the exact-LP theorem bound.

5. PRESERVED CONTROLLED ROOT ABLATION

TABLE XI: Eight historical root pairs: individual widths and their best valid intersection

Case	Unscaled	Scaled	Intersection	Relative (%)	Target
00	0.0908	0.0591	0.0591	2.1807	No
01	0.3359	0.0684	0.0684	1.3056	No
02	0.5289	0.5289	0.5289	9.8157	No
03	1.8087	0.9000	0.9000	4.6349	No
04	0.1311	0.1014	0.1014	1.8478	No
05	0.3039	0.0452	0.0452	0.2781	No
06	28.6138	28.6138	28.6138	100.0000	No
07	15.7973	15.7973	15.7973	100.0000	No

TABLE XII: Historical individual root proposals, not global solutions

Case	Method	LP status	Lower	Upper	Width	LP (s)
00	unscaled	0	2.6281	2.7189	0.0908	0.006
00	scaled	0	2.6530	2.7121	0.0591	0.004
01	unscaled	0	4.9481	5.2840	0.3359	0.050
01	scaled	0	5.1676	5.2360	0.0684	0.072
02	unscaled	0	4.8596	5.3885	0.5289	0.340
02	scaled	0	4.8596	5.3885	0.5289	0.367
03	unscaled	0	17.9216	19.7303	1.8087	9.077
03	scaled	0	18.5169	19.4169	0.9000	27.514
04	unscaled	0	5.3647	5.4959	0.1311	2.182
04	scaled	0	5.3844	5.4858	0.1014	2.032
05	unscaled	0	15.9744	16.2783	0.3039	2.328
05	scaled	0	16.2134	16.2586	0.0452	1.935
06	unscaled	1	0	28.6138	28.6138	30.052
06	scaled	1	0	28.6138	28.6138	30.052
07	unscaled	1	0	15.7973	15.7973	30.078
07	scaled	1	0	15.7973	15.7973	30.077

TABLE XIII: Historical action gaps and savings-ratio extrema

Case	$\underline{d}$	λ_-	λ_+
00	0.003709	-63.6061	539.1951
01	0.000610	-1120.4240	75.1573
02	0.000116	-11155.2283	1228.3552
03	0.000331	-4632.9338	421.3481
04	2.203×10^{-5}	-9787.4264	75078.7316
05	0.000156	-268.5077	10054.3144
06	6.200×10^{-5}	-901.1248	29032.7987
07	7.439×10^{-5}	-22114.4443	3180.9620

None of these eight pairs reached 10^{-3} . Cases 06 and 07 exhausted the LP budget without a usable proposal in either formulation, using zero dual multipliers and the operating-optimal reference for a valid but uninformative $[0, U]$ interval. The other six pairs returned numerical proposals, not solutions of the constrained nonlinear problem. The original protocol amendment after partial observations is retained in the R43 history and is not relabeled as a code-frozen prospective evaluation.

6. ORIGINAL RANDOMIZED SAVINGS AND CONTINUUM EVIDENCE

The following signed savings are $s = L_D - U_R$, where L_D bounds every feasible deterministic policy and U_R is a feasible randomized cost. A positive value proves a policy-class separation; a negative value does not prove that deterministic policies are better. The percentages are computed from the preserved six-decimal display inputs: $100s/(U_R + s)$ and $100s/U_R$. They are approximate display summaries, not new exact certificates. All forty-two rows are retained, including twelve zero-cost rows and three negative signed comparisons. Exactly twenty-seven rows have positive signed savings.

TABLE XIV: Every original signed savings comparison, with normalization

T	Installed	ε	U_R	s	$100s/L_D$	$100s/U_R$
4	defer	0.01	4.730105	0.206584	4.18	4.37
4	defer	0.05	3.890120	1.041948	21.13	26.78
4	neural31001	0.01	0.089277	0.021898	19.70	24.53
4	neural31001	0.05	0.030155	0.044177	59.43	146.50
4	neural31002	0.01	0.125168	0.024289	16.25	19.41
4	neural31002	0.05	0.054026	0.058588	52.03	108.44
4	neural31003	0.01	0.099085	0.022055	18.21	22.26
4	neural31003	0.05	0.037227	0.047069	55.84	126.44
4	occupancy_stress	0.01	4.020718	0.186083	4.42	4.63
4	occupancy_stress	0.05	3.248042	0.941433	22.47	28.98
4	spline129	0.01	0.000000	0.000000	—	—
4	spline129	0.05	0.000000	0.000000	—	—
4	spline33	0.01	0.000000	0.000000	—	—
4	spline33	0.05	0.000000	0.000000	—	—
8	defer	0.01	10.637940	0.172786	1.60	1.62
8	defer	0.05	9.854394	0.896320	8.34	9.10
8	neural31001	0.01	0.402312	0.066330	14.15	16.49
8	neural31001	0.05	0.140458	-0.015946	-12.81	-11.35
8	neural31002	0.01	0.277716	0.058701	17.45	21.14
8	neural31002	0.05	0.059511	-0.015198	-34.30	-25.54
8	neural31003	0.01	0.653902	0.211403	24.43	32.33
8	neural31003	0.05	0.100439	0.244576	70.89	243.51
8	occupancy_stress	0.01	9.365726	0.125235	1.32	1.34
8	occupancy_stress	0.05	8.565982	0.682582	7.38	7.97
8	spline129	0.01	0.000000	0.000000	—	—
8	spline129	0.05	0.000000	0.000000	—	—
8	spline33	0.01	0.000000	0.000000	—	—
8	spline33	0.05	0.000000	0.000000	—	—
12	defer	0.01	15.466905	0.145915	0.93	0.94
12	defer	0.05	14.724122	0.747177	4.83	5.07
12	neural31001	0.01	3.360150	0.133784	3.83	3.98
12	neural31001	0.05	2.023532	0.379048	15.78	18.73
12	neural31002	0.01	2.385046	0.151782	5.98	6.36

T	Installed	ε	U_R	s	$100s/L_D$	$100s/U_R$
12	neural31002	0.05	1.455145	0.309646	17.55	21.28
12	neural31003	0.01	1.010276	0.217318	17.70	21.51
12	neural31003	0.05	0.142107	-0.001624	-1.16	-1.14
12	occupancy_stress	0.01	13.655059	0.044193	0.32	0.32
12	occupancy_stress	0.05	12.877517	0.365025	2.76	2.83
12	spline129	0.01	0.000000	0.000000	—	—
12	spline129	0.05	0.000000	0.000000	—	—
12	spline33	0.01	0.000000	0.000000	—	—
12	spline33	0.05	0.000000	0.000000	—	—

If a feasible lottery costs U_R and deterministic costs are at least L_D , an additional nonnegative implementation charge bounded by F preserves the strict comparison whenever $F < L_D - U_R$. Under a stagewise charge bounded by f , the sufficient condition is $f \sum_{t=0}^{T-1} \beta^t < L_D - U_R$. These are conditional governance budgets in the model's implementation units. No empirical administration-cost estimate or institutional permission to randomize is inferred.

The maximum relative positive-cost randomized widths remain 14.61%, 32.29%, and 34.23% at horizons 4, 8, and 12. Their medians are 5.24%, 14.33%, and 15.75%. The repair-density theorem addresses the same controlled Borel object, but has not been executed here as a new optimizer of these thirty positive-cost rows.

TABLE XV
RETAINED NONLINEAR TWO-STATE INTERVALS AT ALLOWANCE 0.5

T	N	Lower	Upper	Width/upper (%)	Independent check
8	64	0.9438	3.9348	76.01	Yes
16	128	4.1908	7.6644	45.32	No
32	64	6.5389	16.7884	61.05	Yes
32	128	8.1264	12.5226	35.11	Yes (R40)
32	256	8.8718	11.8492	25.13	Yes (R41)

At allowance 0.05 the retained exercise is lower-only: no candidate has a matching all-state feasible upper certificate. The available lower-only summary does not identify every failure as either true policy infeasibility or enclosure conservatism. No such diagnosis is manufactured. This distinction requires an additional candidate-specific certificate or a witnessed violation, neither of which is newly supplied here.

The observable-fiber example remains one fixed-covariate family. Its finest, sixteen-cell interval is approximately $[10.074215, 10.184476]$, with width about 0.110261. Approximately 0.074982 comes from finite endpoint gaps and 0.035279 from partition/value variation. More partition cells cannot remove the former. The new conditional initial-law argument and full Borel-selection proof extend the analytic identity, but do not constitute a new nontrivial-covariate experiment. Exact exogenous service-economy examples are retained as aggregation tests with economically ancillary continuous coordinates.

6.1. Nested fiber work and primitive errors

The cell reference is its left endpoint on $z \in [0, 1]$. In the original example $r_z = (1 + z)r$, $g_z = (1 + z)g$, $k_z = (1 + z/2)k$, and the finite regime matrix is independent of z . For M equal cells, valid uniform primitive errors are $d_r = \max |r|/M$, $d_k = \max k/(2M)$, $d_g = \max |g|/M$,

and $\alpha = 0$ in the finite regime matrix norm. The following numbers are evaluated on the exact stored base model. They are generic error inputs, not the sharper monotonicity partition term used to obtain the interval above.

TABLE XVI: Cellwise primitive errors in the original fiber example

Cells	d_r	d_k	d_g	α
2	0.50000000	0.50000000	0.25000000	0
4	0.25000000	0.25000000	0.12500000	0
8	0.12500000	0.12500000	0.06250000	0
16	0.06250000	0.06250000	0.03125000	0

TABLE XVII: Nested fiber width and the actual needed endpoint work

Cells	Endpoints	Width	Finite part	Partition part	LP (s)	Build/arith. (s)	Check (s)
2	3	0.342810	0.056917	0.285893	0.0762	0.1548	0.1625
4	5	0.208049	0.066491	0.141558	0.1291	0.2591	0.2708
8	9	0.142796	0.072193	0.070603	0.2293	0.4684	0.4854
16	17	0.110260	0.074981	0.035279	0.4326	0.8939	0.9152

The work sums the unique stored endpoints needed for each nested partition. These are not four freshly executed mesh experiments, and their times must not be added as independent total work. The finest construction uses seventeen endpoint solves. No tighter finite search or nonidentity label evolution is newly executed in this calculation.

7. VERIFICATION, REPRODUCIBILITY, AND ANALYTIC CONTRACTS

The present recheck covers all 32 complete custom trees, all 16 repaired SCIP candidate policies, and all 42 scaling root proofs. All 90 passed. Eleven frozen scientific-source identities and sixteen primitive identities were also matched. The local interpreter is Python 3.13.5; these rechecks do not remeasure the original solver benchmark on Python 3.13.15. The global replay report can resume previously completed checks, so its coordinator elapsed time must not be interpreted as a fresh single-run timing of every object.

The archived adversarial suite is kept separately from the new stopping-modulus tests. It checks mutated proof structures and selected support identities. Source and primitive hashes establish identity, not correctness. Arithmetic readers rebuild finite recursions, constraints, cover structure, and residual bounds; they do not machine-prove measurable selection, density, analytic aggregation, or the economic assumptions. The machine-readable `ASSUMPTIONS.json` records these boundaries for every headline result.

To regenerate the tables without an optimizer, run `replication/tabulate.py` in the `R45` directory using the supplied workspace. To replay certificates, run `replication/recheck.py` and `replication/recheck_scaling.py`. To reproduce the exact new property test and precision calculations, run `replication/check_k_search_modulus.py`. The plotting script uses only the archived traces and scaling rows. All command paths are supplied in the root review guide.

8. PRESERVATION AND DOCUMENT MAP

The principal article contains the model, central results, complete proofs of the new stopping and density results, the global comparison, and the economic conclusions. This numerical supplement contains all newly tabulated rows, including unsuccessful outcomes. The response document addresses all seventeen blocking findings and all thirty additional comments. The historical volume reproduces the full R43 article and supplement together with the inherited R42 historical volume. Their original PDFs and source files remain available as distinct unchanged files.

The preservation manifest records 461 inherited files in the supplied working package. The proposed repository update is add-only under a new R45 directory and new R45 entry points. It neither deletes other branches nor rewrites the larger remote historical archive. The local package is not advertised as a new complete clone of every historical branch or as containing the entire remote 349-page annex. Unmaterialized remote history remains untouched by the add-only publication procedure.